

Tuned machine learning modeling tools and genetic algorithm for anionic dye adsorption on biomass-based activated carbons

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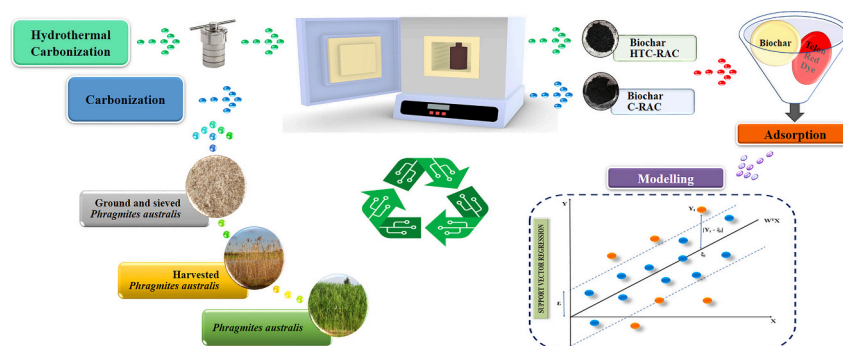
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HIGHLIGHTS

- Biochars were produced by carbonization and hydrothermal carbonization methods.
- Pseudo-second-order and Langmuir models presented the best conformity.
- The adsorption capacity of HTC-RAC was significantly higher than the others.
- GPR, as the best process simulation, produced consistent and reliable results.
- Input levels to produce the best adsorption were obtained by genetic algorithm.

GRAPHICAL ABSTRACT



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ABSTRACT

In this study, the adsorption behaviors of *Phragmites australis*-based biochars and commercial activated carbon were compared for Telon Red dye removal. *Phragmites australis*-based biochars, C-RAC and HTC-RAC were produced by carbonization and hydrothermal carbonization methods, respectively. The adsorption capacities of CAC, C-RAC, and HTC-RAC were calculated as 262.66, 255.77, and 806.33 mg/g, respectively. Thermodynamically, ΔG° showed that the adsorption behavior for all adsorbents occurs spontaneously. ΔH° values showed that the dye adsorption on CAC and HTC-RAC were controlled by a physical mechanism, unlike C-RAC. The ΔS° values indicated that Telon Red molecules were randomly distributed on C-RAC during the adsorption processes, in contrast to CAC and HTC-RAC. Moreover, Gaussian Process Regression, Support Vector Regression, and Neural Network Regression with different strengths were used to model adsorption behaviors. From the optimal outcomes achieved through hyperparameter tuning, Gaussian Process Regression was the most effective modeling tool producing the closest-to-reality simulation. According to the MAPE criterion, an error rate of 0.8134 % for C-RAC, 0.9324 % for CAC, and 0.7924 % for HTC-RAC were obtained in the simulations. The genetic algorithm was used to optimize these reliable and valid simulations to have the highest adsorption capacity with nearly desirability of 100 %. Finally, the adsorption capacities of all three adsorbents were compared statistically, and it was concluded that the adsorption capacity of HTC-RAC was significantly higher than the others.

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1. Introduction

The textile industry is one of the oldest and largest industries in the world. The changing and developing living conditions have led to the diversification of textile products, thus modernization and rapid industrialization [1]. According to World Bank data, the textile industry, which contributes greatly to the global economy, is responsible for 20 % of the world's industrial water pollution. Large amounts of water and various chemicals are used in the textile industry. According to the US Environmental Protection Agency, 100–200 L of water is required to process 1 kg of fabric [2,3]. During wet processing in the textile industry, large amounts of wastewater containing dye bath carriers, salts, heavy metals, emulsifiers, dispersing/levelling agents and non-biodegradable organic compounds (especially dyes) are generated [4,5]. Many physical, chemical and biological treatment methods such as coagulation-flocculation [6,7], biodegradation [8,9], advanced oxidation methods [10–12], photocatalytic degradation [2,13], electrochemical methods [14,15], membrane filtration [16,17] and adsorption [18,19] have been investigated to remove dye molecules from textile industry wastewater. Moreover, each method's cost, efficiency, and environmental impact were evaluated [20].

Adsorption is a prominent method for removing dyes from wastewater due to its high efficacy in removing various contaminants, low cost, ease of usage and design, and insensitivity to harmful compounds [21]. Alternative natural, affordable, and efficient adsorbents to activated carbon, which is useful in dye adsorption but is an expensive adsorbent for large-scale treatment systems, have been investigated [1,5,22–27]. Biochar is a carbon-rich material made by thermally digesting various waste biomass such as forestry, industrial, and agricultural waste in an oxygen-poor or no-oxygen environment [28,29]. High-temperature pyrolysis, gasification, hydrothermal carbonation, and baking of biomass materials are some of the biochar preparation techniques [30]. Physical and chemical modifications are applied to improve its adsorption and activation ability in addition to the physical properties of biochar. Physical modification optimizes the crystalline pore structure and thus increases the hydrophilicity and performance of biochar. Chemical modification such as acid-base, oxidant and metal modifications can also provide biochar with magnetism as well as increase the diversity, amount and surface area of functional groups [31–33]. Because of its fundamental benefits such as its larger surface area and porosity, low cost of generation, variety of surface functional groups, ease of surface modification applications, remarkable hydrophobicity, and aromaticity, biochar gets attention as an adsorbent [34]. Therefore, biochar based on the applied method offers similar properties to activated carbons, biosorbents, and other adsorbents and is ten times more economical than them [29]. Biochars such as *Opuntia ficus indica* [35], chicken bone [36], lychee seed [37], tapioca peel waste [38], tea residue [39], *Eichhornia crassipes* and *Phragmites australis* stems [40], cotton stalk [41] obtained from biomasses by different thermal methods, have been the focus of research for the removal of dyestuffs from wastewater.

The right mathematical modeling tool and accurate predictions for an adsorption process are crucial. In this regard, various modeling tools are used in the literature. Batra and Chen constructed a machine learning-based model to accurately and instantly classify metal–organic frameworks (MOFs) [42]. Aftab et al. introduced ML models to predict the adsorption breakthrough curves of four heavy metals (Cd, Cu, Pb and Zn) over low cost multi-metal binding biosorbent [43]. Zhao et al. presented a study involving an application of machine learning in predicting the adsorption capacity of organic compounds onto biochar and resin [44]. Wu et al. proposed a model based on neural networks to accurately predict methane adsorption isotherms of metal–organic frameworks [45]. Yang et al. proposed the use of machine learning algorithms in MOFs for gas adsorption and separation [46].

However, the modeling tools presented have been solely utilized to obtain predictions related to the process. Also, the mathematical

modeling of the adsorption process involves simulating the transaction. The better a mathematical model produces predictions about the actual process, the more accurate the simulation of the process becomes. A well-simulated model or a good enough mathematical model can be used to improve the process without the need for new experimental designs. This is much more valuable than simply producing good predictions. In many studies, the focus has been on prediction accuracy and has been determined subjectively rather than optimizing/tuning the hyperparameters of modeling tools. However, model hyperparameters play a key role in creating an effective modeling tool to provide a high-quality simulation of the process.

This study aimed to compare and evaluate the adsorption behaviors of *Phragmites australis*-based activated carbons generated by carbonization and hydrothermal carbonization methods with commercial activated carbon for Telon Red dye removal. Moreover, three different machine learning models with different strengths were used to model and simulate behaviors of these adsorption: *Gaussian Process Regression*, *Support Vector Regression*, and *Neural Network Regression*. Among these three machine learning models, the model with the most desired features was identified to create a simulation of the process. This study's objectives and distinguished aspects compared with the others are as follows: (i) using kinetic and isotherm models to understand the Telon Red adsorption mechanism; (ii) using thermodynamic parameters to determine the suitability of the adsorption; (iii) comparing the Telon Red dye removal performances of all adsorbents; (iv) proving the predictive capabilities of 3 ML-based modeling tools; (v) decision-making the simulation of the Telon Red adsorption mechanism by tuning the hyperparameter of the ML-based modeling tools; (vi) evaluating the reliability of the specified simulation using the rerun principle; and (vii) determining input parameter values, by genetic algorithm, that will produce the desired/expected outputs through optimization of the process simulation.

2. Materials and methods

2.1. Adsorbents

The source of raw material used in the production of biochar was *Phragmites australis* (reed). Approximately 25 % of the reeds in the delta are cut every year. *Phragmites australis* was provided by farmers who made their living by trading cane in the Kızılırmak Delta, located in the Bafra plain in Samsun city. After the dried reed, which was obtained directly from the farmer in the field, was brought to the laboratory, it was cut into 10–20 cm size and stored in polyethylene containers at +4 °C until the experimental studies. For biochar production, reed samples cut to 10–20 cm in size were ground and sieved, and samples with a size of 0.425–0.600 mm were used in the carbonization and hydrothermal carbonization processes.

In the study, the adsorption capacities of biochars obtained from two different carbonization processes will be compared with commercial activated carbon (CAC). Activated carbon produced by activating charcoal had a molecular mass of 12.01 g/mol (Merck).

In the carbonization process, the sieved biomass was mixed with 99.0 % purity NaOH in a ratio of 1:3 and placed in a reactor made of stainless steel with an inner diameter of 4.5 cm, an inner depth of 8.30 cm, and a wall thickness of 0.5 cm. The reactor was tightly closed to prevent air ingress and set in a muffle furnace. The carbonization process was carried out at 600 °C for 1 h. The pH value of the charcoal obtained after the process was around 12.50–13.50. Therefore, the charcoal was placed on filter paper and washed with distilled water using a vacuum pump to neutralize the pH value (6–7) and dried in the oven for 24 h at 105 °C [47]. The activated charcoal obtained after drying and pounding in a mortar was called "Reed Activated Carbon (C-RAC)".

In the hydrothermal carbonization process, the sieved biomass was mixed with distilled water to 5 % by weight and placed in a reactor made

of paraphenyl phenol with an inner diameter of 3.0 cm, an inner depth of 7.0 cm, and a wall thickness of 0.5 cm. The reactor was put in a hydrothermal reactor made of stainless steel with an inner diameter of 6.5 cm, an inner depth of 14.0 cm, and a wall thickness of 0.5 cm. The hydrothermal carbonization process was performed at 200 °C for 5 h in a muffle furnace. The solid-liquid mixture was separated and hydro-coke dried in the oven for 24 h at 105 °C. Then, similar to the carbonization process, the hydro-coke was mixed with 99.0 % purity NaOH in a ratio of 1:3 and placed in the reactor. The hydro-coke was activated at 600 °C for 1 h in a muffle furnace. The activated hydro-coke was placed on filter paper and washed with distilled water using a vacuum pump to neutralize the pH value and dried in the oven for 24 h at 105 °C [48,49]. The activated hydro-coke was pounded in a mortar and coded "Reed Activated Carbon (HTC-RAC)". SEM images of C-RAC and HTC-RAC are given in Fig. 1.

As seen in Fig. 1a, the surface of the C-RAC obtained from the carbonization process was almost non-porous. Although the formation of pores in the biochar prepared by the hydrothermal method is not expected due to the condensation of the volatile matter [49], a large number of pores, large and small, were formed (Fig. 1b). Hydrothermal carbonization process changed the surface properties of the adsorbent by affecting the lignin, cellulose, hemicellulose composition of the cell wall. Thus, the NaOH molecule easily penetrated the hydro-coke and created many pores.

Fig. 2 presents XRD patterns and FTIR spectra of C-RAC and HTC-RAC. Fig. 2a shows the XRD patterns of C-RAC and HTC-RAC samples in the 2θ range of 5° to 90°. The broad peak with a low intensity appearing in the X-ray pattern demonstrates the amorphous structure of C-RAC and HTC-RAC which is evidence of a high oxidation degree caused by the activation process. The XRD peak at 2θ ~ 23° is characteristic for amorphous carbon, and the peak located at 2θ ~ 43° corresponds to graphite crystallites. The elevated background signal between 10° and 20° may be ascribed to the existence of micropores in the samples. Eventually, the X-ray beam is scattered by these micropores, leading to an increased background signal. Fig. 2b shows FTIR spectra of C-RAC and HTC-RAC adsorbents. The peaks at 2901 and 2989 cm⁻¹ in C-RAC adsorbent correspond to aliphatic C—H stretching originating from cellulose and hemicellulose residues, while these peaks are not observed in HTC-RAC adsorbent obtained by hydrothermal carbonization method. The bending vibrations of 1395 and 1498 cm⁻¹ originate from aliphatic hydrocarbons and peak intensity decreased after hydrothermal carbonization process. The C—O stretching represented by the absorbance at 1242 cm⁻¹ indicates ester and phenol compounds. The peak at 1065 cm⁻¹ indicating C—O—C stretching in ether and polysaccharides is associated with cellulose and hemicellulose residues and disappeared

after hydrothermal carbonization process.

2.2. Adsorbate

Telon Red, an acidic and reactive dye, is a negatively charged, water-soluble anionic dye used in textile industries such as wool, nylon, and silk. Telon Red is a dark red dye with strong water retention properties. The molecular structure of Telon Red dye, whose molecular formula is C₂₂H₁₆N₃NaO₆S₂, is given in Fig. 3 [50].

2.3. Adsorption experiments

The batch system adsorption experiments were carried out to evaluate the effects of initial dye concentration, initial pH value, adsorbent dose, contact time, and temperature changes on Telon Red dye removal capacities of C-RAC, HTC-RAC, and CAC. The experiments based on initial dye concentration, initial pH value, adsorbent dose, and contact time were conducted in a stirrer water bath (JULABO SW22) and 150 rpm. pH value of adsorbate was adjusted by 1 M H₂SO₄ and NaOH. The supernatant of the mixture was separated by centrifugation at 6000 rpm. Telon Red concentration in the supernatant was measured with UV-Vis Spectrophotometer (HACH LANGE DR6000) at a wavelength of 500 nm. Telon Red adsorption capacities (*q*) and removal efficiencies (*RE*) of the adsorbents were calculated by Eqs. 1 and 2, respectively.

$$q = \frac{(C_0 - C_t)V}{M} \quad (1)$$

$$RE (\%) = \frac{C_0 - C_t}{C_0} \times 100 \quad (2)$$

where, *C*₀ is the initial Telon Red concentration (mg/L), *C*_{*t*} is Telon Red concentration in *t* time, *M* is the adsorbent amount (g), *V* is the waste-water volume (L), and *q* is the amount of Telon Red adsorbed per gram adsorbent.

2.4. Kinetic evaluations of adsorption process

In the pseudo-first-order reaction model, it is assumed that the saturation of the adsorbent and physical adsorption are directly proportional to the unused areas in the adsorbent [51]. The model's linear equation is seen in Eq. 3. In the pseudo-second-order reaction model (Eqs. 4 and 5), it is assumed that there is an electron exchange between adsorbate and adsorbent; therefore, chemical adsorption is dominant in the process [52]. The Elovich model, which the linear equation is given in Eq. 6 assumes that chemical adsorption occurs on adsorbents with

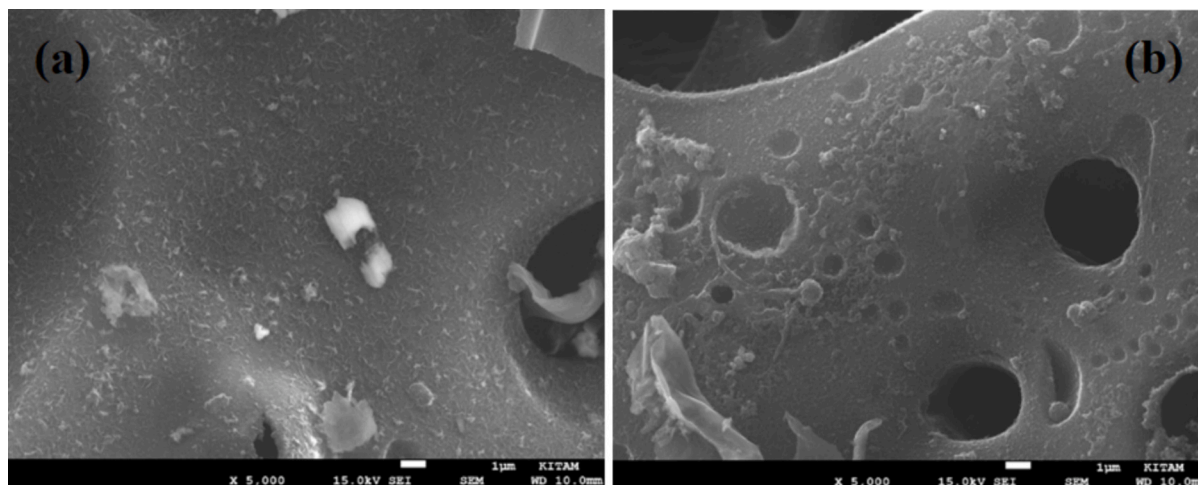


Fig. 1. SEM image of C-RAC (a) and HTC-RAC (b).

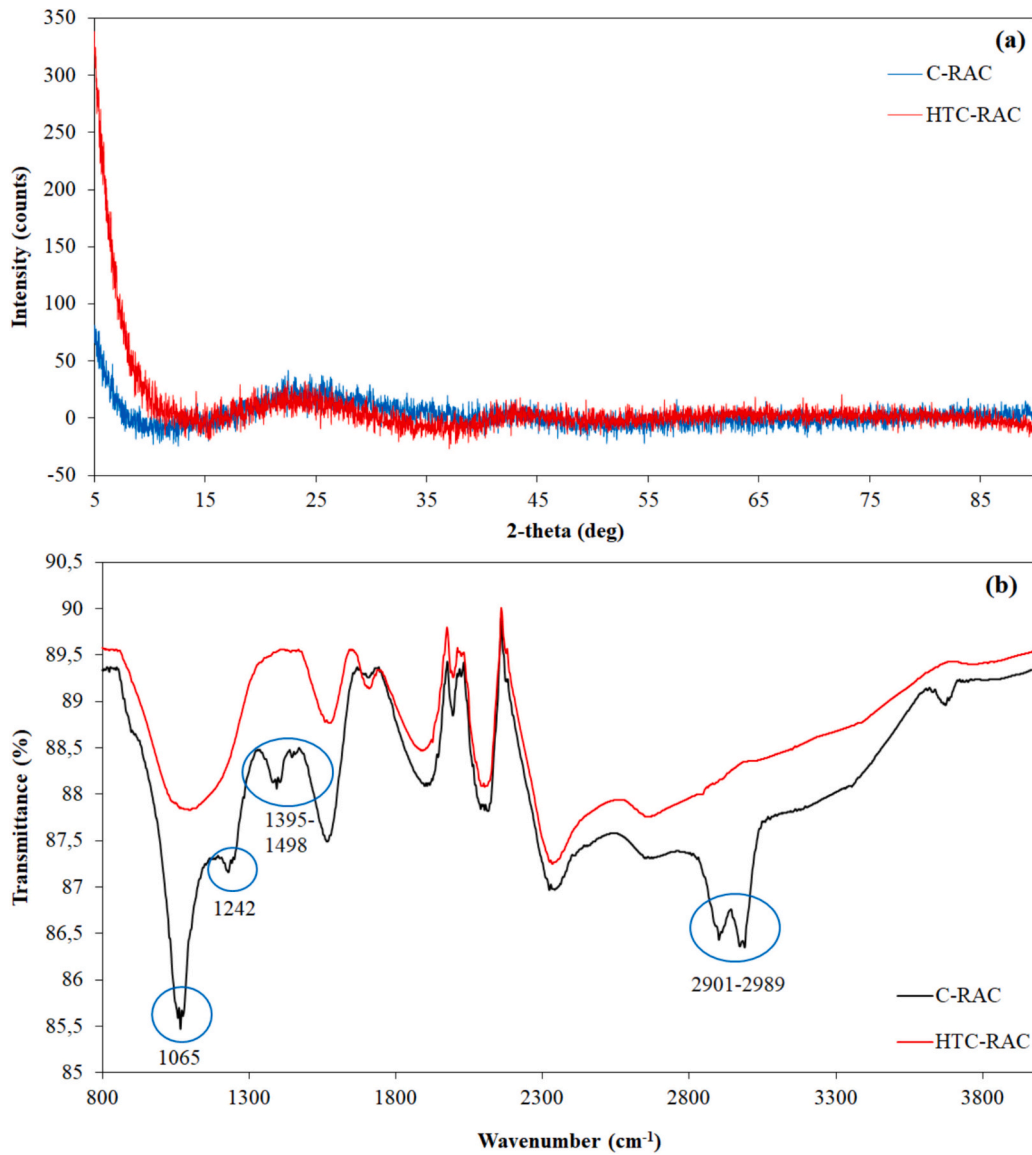


Fig. 2. XRD patterns (a) and FTIR spectra (b) of C-RAC and HTC-RAC.

heterogeneous structure (Eq. 6). The intra-particle diffusion model describes the transfer of adsorbate to adsorbent [53].

$$\ln(q_e - q_t) = \ln q_e - k_1 t \quad (3)$$

$$\frac{t}{q_t} = \frac{1}{k_2 \cdot q_e^2} + \frac{t}{q_e} \quad (4)$$

$$h = k_2 \cdot q_e^2 \quad (5)$$

$$q_t = \frac{1}{\beta} \ln(\alpha \beta) + \frac{1}{\beta} \ln t \quad (6)$$

$$q_t = k_{id} t^{1/2} + C \quad (7)$$

where, t is contact time (min), k_1 is the rate coefficient of Lagergreen (1/min), q_t and q_e , are the adsorption capacities (mg/g) of t time and the equilibrium time, respectively. k_2 is the adsorption velocity constant of the pseudo-second-order reaction model (g/mg min), h is the inlet adsorption rate (mg/g min), α is the chemical adsorption rate (mg/g min), β is a constant related to the surface coverage and activation energy for chemisorption (g/mg). In Eq. 7, k_{id} is the rate constant (mg/g h^{1/2}

²), C is a constant related to the boundary layer dimension and t is contact time (h).

2.5. Isotherms evaluations of adsorption process

Mathematical formulas of isotherm models are used to calculate the other characteristics, like adsorption capacity, and for predicting the mechanism of adsorption. A number of adsorption models (homogeneous/heterogeneous or monolayer/multilayers) have been created to fit experimental data and predict the mechanism of adsorption [54]. The Langmuir isotherm model describes that the adsorption process is energy-equivalent, monolayer, and homogeneous on the adsorbent surface [55]. The linear equation of the model is given in Eq. 8. The linear equation of Freundlich isotherm model is given in Eq. 10. The Freundlich isotherm explains the exponential distribution of active sites and their energies in the adsorbent, thus the adsorption process has a heterogeneous structure [56]. Tempkin model explains that the adsorption process is homogeneous on the adsorbent surface and the adsorption heat of molecules reduces linearly (Eq. 11) [19].

Here, C_0 and C_e are inlet and final concentrations of Telon Red (mg/L), respectively. q_e and q_m is the equilibrium and maximum adsorption

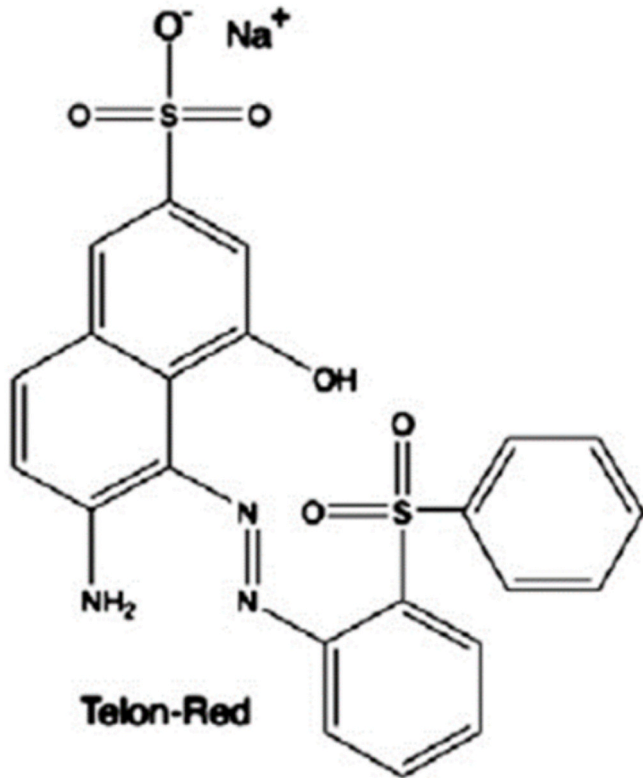


Fig. 3. The chemical structure of Telon Red dye.

capacities of adsorbents (mg/g), respectively. K_L is a Langmuir model constant (L/mg), and R_L explains the interaction between the adsorbate and adsorbent (dimensionless). K_F is the constant of Freundlich model (mg/g), n gives the adsorption intensity (L/g). R is the universal gas constant (8.3145 J/mol K), T is the temperature of adsorbate (K), b is a constant of the adsorption heat (J/mol), and K_T is the equilibrium binding coefficient.

$$\left(\frac{C_e}{q_e}\right) = \left(\frac{1}{q_m K_L}\right) + \left(\frac{C_e}{q_m}\right) \quad (8)$$

$$R_L = \frac{1}{(1 + K_L C_0)} \quad (9)$$

$$\log q_e = \log K_F + \left(\frac{1}{n}\right) \log C_e \quad (10)$$

$$q_e = \frac{RT}{b} \ln K_T + \frac{RT}{b} \ln C_e \quad (11)$$

2.6. Machine learning

Machine learning (ML) algorithms have shown remarkable success in solving regression problems where the goal is to predict continuous values based on input data. Mirroring the way humans learn, it enables systems to automatically recognize patterns, make decisions, and adapt to new information. From this perspective, ML is an automatic learning process that processes the data sets [57]. A supervised ML algorithm learns directly from the data and learns to connect one or more inputs to one or more outputs to constitute predictions related to new data [58,59]. In this study, three ML-based modeling tools with the distinct strengths were selected to simulate the Telon Red adsorption mechanism: Gaussian Process Regression, Support Vector Regression, and Neural Network Regression.

2.6.1. Gaussian Process Regression

Gaussian Process Regression (GPR) is a powerful machine learning algorithm used for regression tasks. Unlike conventional regression techniques that often rely on predefined parameterized relationships between input variables and output targets, GPR employs a non-parametric Bayesian approach. This method models the entire distribution of potential functions that could describe the relationship between inputs and outputs. Unlike most of the widely used supervised ML algorithms, which aim to determine precise values for each parameter within a function, the Bayesian approach instead derives a probability distribution over all possible values [60].

Let the independent and dependent variable data set be $\{(x_i, y_i); i = \overline{1, n}\}$, where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$ from an unknown distribution. A GPR model addresses the issue of predicting the value of a response variable y , given the input vector x . A linear regression model is of the form (Eq. 12);

$$y = x^T \beta + \varepsilon \quad (12)$$

where $\varepsilon \sim \text{Normal}(0, \sigma^2)$. Also, σ^2 is the error variance and β is the vector of model coefficients. Both are estimated from the data. A GPR model explains the response by constructing *latent variables*, $f(x_i)$, $i = \overline{1, n}$, based on a Gaussian Process (GP), and explicit *basis functions*, h . Basis functions transform the inputs x into a p -dimensional feature space. Moreover, the covariance function of the latent variables captures the patterns of the response. GP consists of any finite number of sets of random variables having a joint Gaussian distribution. Here, if $\{f(x); x \in \mathbb{R}^d\}$ is a GP, then given n observations $x_1, x_2, \dots, x_n$ the joint distribution of the random variables $f(x_1), f(x_2), \dots, f(x_n)$ is Gaussian. GP is defined by the parameters; $m(x)$ which is the mean function and $k(x, x')$ which is the covariance function. That is, if $\{f(x); x \in \mathbb{R}^d\}$ is a GP;

$$E(f(x)) = m(x) \quad (13)$$

$$\text{Cov}[f(x), f(x')] = E[\{f(x) - m(x)\} \{f(x') - m(x')\}] = k(x, x') \quad (14)$$

A GPR model can be represented as given in Eq. 15.

$$y = h(x^T) \beta + f(x) + \varepsilon \quad (15)$$

here $f(x) \sim GP(0, k(x, x'))$. That is, $f(x)$ are from the GP with a zero mean and with $k(x, x')$ covariance function. Also, $h(x)$ are a set of basis functions that project the original independent variables vector x in d -dimension into a novel independent variables vector in p -dimension. Moreover, β is p -by-1 coefficients vector for basis function. In this case, an instance of response variable y is modelled as in Eq. 16;

$$P(y_i | f(x_i), x_i) \sim \text{Normal}(y_i | h(x_i)^T \beta + f(x_i), \sigma^2) \quad (16)$$

The latent variables $f(x_i)$ introduced for each observation x_i makes the GPR model nonparametric. This model is given in Eq. 17 in a vector form.

$$X = \begin{pmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{pmatrix}; y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}; H = \begin{pmatrix} h(x_1^T) \\ h(x_2^T) \\ \vdots \\ h(x_n^T) \end{pmatrix}; f = \begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{pmatrix} \quad (17)$$

The joint distribution of latent variables $f(x_1), f(x_2), \dots, f(x_n)$ in the GPR model is given as in Eq. 18.

$$P(f|X) \sim \text{Normal}(f | 0 + K(X, X)) \quad (18)$$

where $K(X, X)$;

$$K(X, X) = \begin{pmatrix} k(x_1, x_1) & k(x_1, x_2) & \dots & k(x_1, x_n) \\ k(x_2, x_1) & k(x_2, x_2) & \dots & k(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ k(x_n, x_1) & k(x_n, x_2) & \dots & k(x_n, x_n) \end{pmatrix} \quad (19)$$

2.6.2. Support Vector Regression

Support Vector Regression (SVR) is a modification of support vector machines, as the name suggests, is a regression algorithm that is used to solve both linear and non-linear regression problems. In common regression algorithms, the basic idea is to minimize the error, while in SVR, it is to limit the error within a certain threshold. That is, SVR aims to approximate the best value within a given margin called ε -tube. The model to be estimated for a linear regression problem can be written as follows:

$$f(x) = x'\beta + b \quad (20)$$

This model requires estimating $f(x)$ with the minimal norm value of (β, b) . This is also a convex optimization problem to minimize the following function, subject to all residuals having a value less than ε and formulated in Eq. 22.

$$J(\beta) = \frac{1}{2} \beta' \beta \quad (21)$$

$$\forall i : |y_i - x'_i \beta - b| \leq \varepsilon \quad (22)$$

No such function $f(x)$ may exist to satisfy these constraints for all data points. To deal with, in that case, infeasible or unsatisfied constraints, slack variables ξ_i and ξ_i^* are introduced for each data point. These slack variables allow regression errors to exist up to the value of ξ_i and ξ_i^* , yet still satisfy the required conditions. By including slack variables, the formed objective function can be given as in Eq. 23 subject to constraints given in Eq. 24.

$$J(\beta) = \frac{1}{2} \beta' \beta + C \sum_{i=1}^n (\xi_i + \xi_i^*) \quad (23)$$

$$\forall i : y_i - x'_i \beta - b \leq \varepsilon + \xi_i \quad \forall i : x'_i \beta + b - y_i \leq \varepsilon + \xi_i^* \quad (24)$$

$$\forall i : \xi_i^* \geq 0 \quad \forall i : \xi_i \geq 0 \quad (25)$$

The C parameter, referred to as the box constraint, is a positive numeric value regulating the penalty applied to instances positioned beyond the epsilon margin (ε). The optimization problem defined in Eqs. 23 and 24 is commonly solved in a form using Lagrange multipliers given in Eq. 25.

$$L = \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*) - \sum_{i=1}^n \alpha_i (\varepsilon + \xi_i - y_i + \langle \beta, x_i \rangle + b) - \sum_{i=1}^n \alpha_i^* (\varepsilon + \xi_i^* + y_i - \langle \beta, x_i \rangle - b) - \sum_{i=1}^n (\eta_i \xi_i + \eta_i^* \xi_i^*) \quad (25)$$

where $\forall i : \alpha_i, \alpha_i^*, \eta_i, \eta_i^* \geq 0$. Moreover, in the optimal condition, the partial derivatives of L with respect to these variables ($\forall i : \alpha_i, \alpha_i^*, \eta_i, \eta_i^*$) are equal to zero.

Thus, the form of f given in Eq. 20 can be re-formulated in a general form as follows.

$$f(x) = \sum_{i=1}^n (\alpha_i - \alpha_i^*) k(x_i, x) + b \quad (26)$$

Here, the kernel function $k(x_i, x)$ determines whether the model is linear or non-linear. Three commonly used basic kernel functions are

Linear ($k(x_i, x) = x'_i x_k$), Gaussian ($k(x_i, x) = \exp(-\|x_i - x_k\|^2)$ -non-linear), and polynomial ($k(x_i, x) = (1 + x'_i x_k)^q, q \in \{2, 3, \dots\}$ -non-linear) functions.

2.6.3. Neural Network Regression

In the modern sense, neural networks, Multilayer Perceptron Neural Networks, suggested by Werbos [61] to solve non-linear problems and improved by Rumelhart et al. [62], have been widely used in regression problems. In neural network regression, the input data is fed into the network, which then processes the information through multiple layers of neurons to produce an output prediction. The network learns to map input features to output targets by adjusting the weights and biases of the connections between neurons during training (Fig. 4).

where, P is the input vector. The number of inputs is R . $IW^{1,1}$ is the *Input Weight* matrix having a source 1 (second index — from the input layer) and a destination 1 (first index — to the first hidden layer) and b^1 is a bias vector for the first hidden layer. The number of neurons in the first hidden layer is S^1 . a^1 is the output vector for the first hidden layer and also is the input vector for the second hidden layer at the same time. A constant input 1 is fed to the biases for each neuron in the first hidden layer. $LW^{2,1}$ is the *Layer Weight* matrix having a source 1 (second index — from the first hidden layer) and a destination 2 (first index — to the second hidden layer) and b^2 is a bias vector for the second hidden layer. The number of neurons in the second hidden layer is S^2 . Moreover, $LW^{3,2}$ is the *Layer Weight* matrix having a source 2 (second index — from the second hidden layer) and a destination 3 (first index — to the output layer) and b^3 is a bias vector for the output layer. The number of neurons in the output layer is S^3 . In this study, S^3 , the number of neurons in the output layer, is taken as 1.

2.6.4. Genetic algorithm

A distinguishing and superior aspect of this study compared to existing ones is the optimization of the modeling tool, which was determined as a process simulation, using a genetic algorithm (GA), in addition to the tuning of the ML algorithm's hyperparameters. The optimal input parameter values that will provide the highest adsorption performance were determined using a genetic algorithm (GA). Gas, developed by John Holland [63] and improved by De Jong [64], works by evolving a population of candidate solutions over multiple generations. Each individual in the population, holding the candidate solution, is subjected to the process of natural selection, crossover, and mutation to produce new individuals that are potentially better suited to the problem at hand [65].

3. Results and discussions

The performances of CAC, C-RAC, and HTC-RAC in the removal of Telon Red dye were compared with batch adsorption experiments. The effects of adsorbent dose, initial pH value, contact time, initial dye concentration, and temperature changes were evaluated to determine the optimum adsorption capacity.

3.1. Performance evaluation of adsorbents

pH level of the adsorbate affects the adsorption process. Because the pH value plays a crucial role in changing the ionization levels of adsorbents and adsorbates [66]. In this study, the pH value of the Telon Red dye solution is 6.41. The initial pH value of the dye solution was

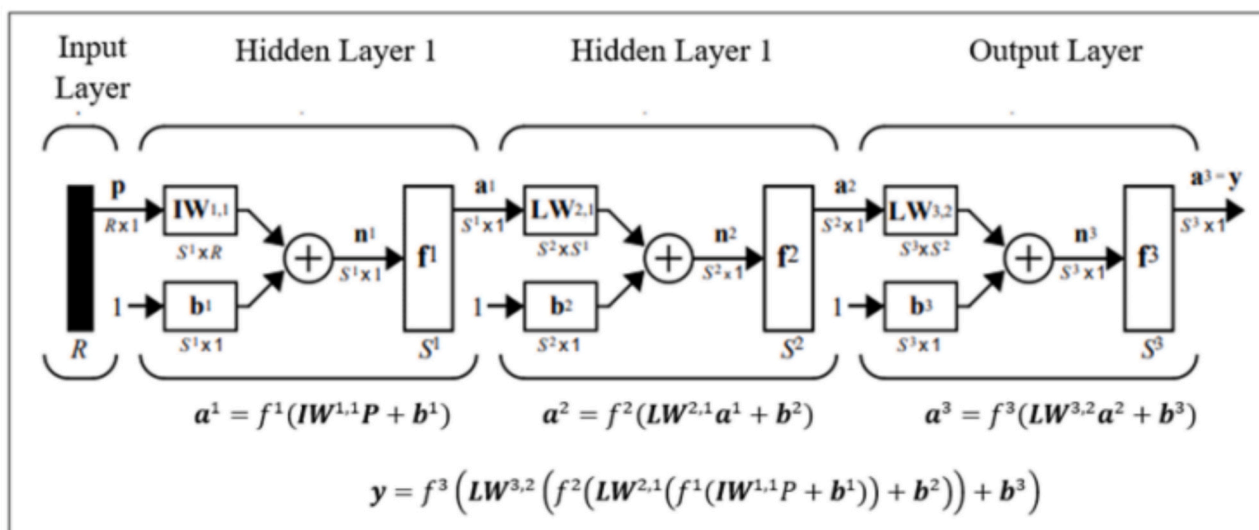


Fig. 4. Neural network structure with one input, two hidden, and one output layers.

adjusted to 3, 5, 6, 7, 8, 9, and 11 to determine the pH value at which the optimum Telon Red adsorption would be achieved. The studies were performed isothermally at 150 rpm in a 20 °C stirrer water bath for 1 h. Adsorbent doses of 0.35 g/L for CAC and C-RAC and 0.1 g/L for HTC-RAC were used to remove 100 mg/L of Telon Red. The effects of pH

changes with the comparative graph created for all adsorbents are shown in Fig. 5a. As is seen in Fig. 5a, the adsorption capacity of CAC, C-RAC, and HTC-RAC decreased as the pH of the dye solution increased and remained almost constant in the pH range of 5–7. This case occurred because the Telon Red dye creates additional OH^- ions and comes into

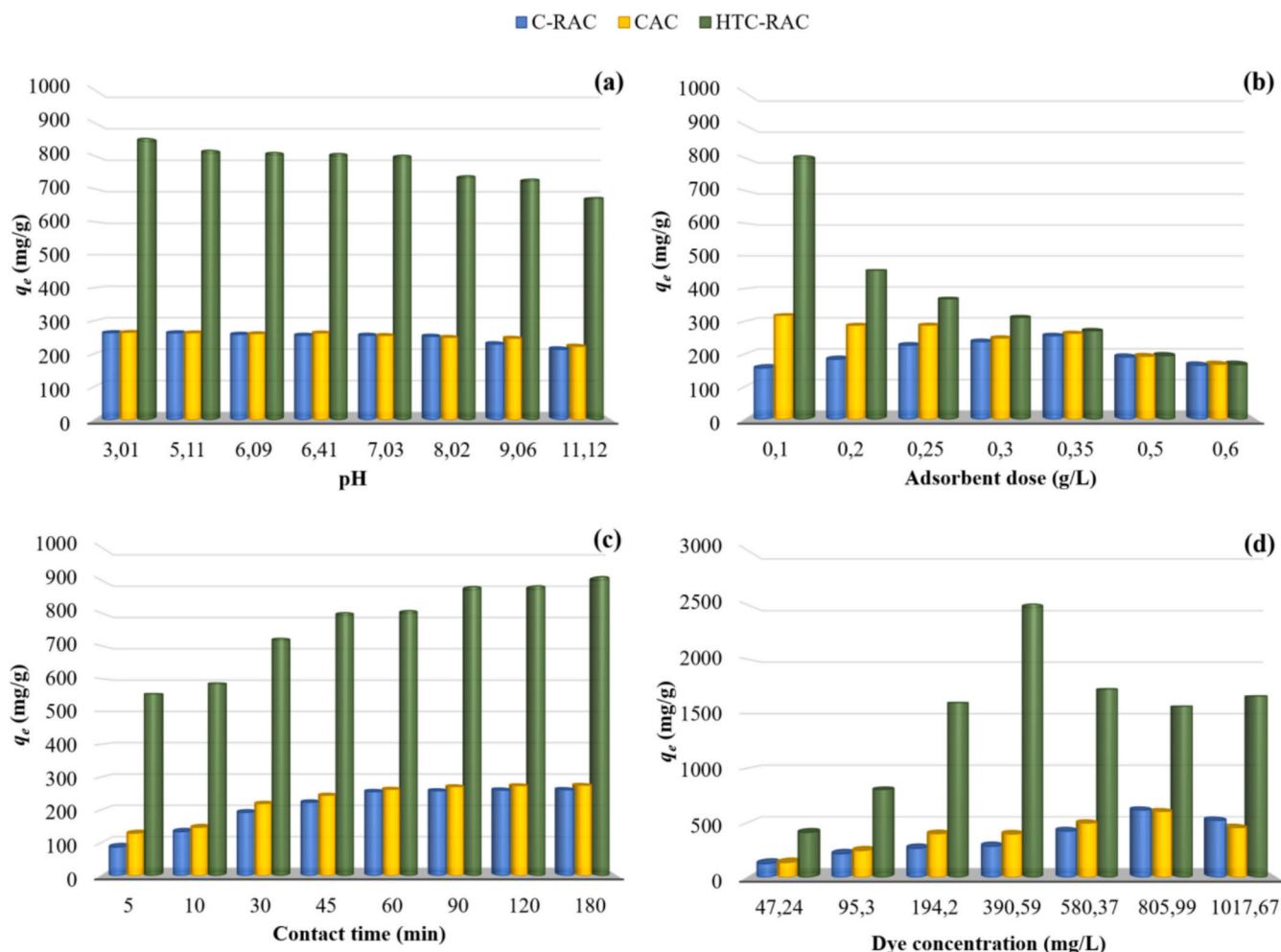


Fig. 5. The effects of pH (a), adsorbent dose (b), contact time (c) and initial dye concentration (d) on Telon Red adsorption capacities of adsorbents.

contact with the positively charged surface of CAC, C-RAC, and HTC-RAC. Telon Red is a negatively charged (anionic) dye and provides negative ions in solutions. Thus, above pH 7, the adsorption capacity was lower due to the electrostatic repulsion between Telon Red ions and the negatively charged surfaces of CAC, C-RAC, and HTC-RAC. Moreover, the number of negative charges on the adsorbent surface increased when the dye solution was basic. This also did not support anionic Telon Red dye adsorption due to electrostatic repulsion. In some cases, solutions with high OH^- ions reduce the adsorption capacity of the system due to friction between dye anions and positively charged ions for adsorption sites. The adsorption process became more effective as the pH shifted to the acidic side. The effects of pH were similar for both C-RAC and CAC. At the lowest pH value, adsorption capacities of CAC and C-RAC were 264.52 and 263.70 mg/g, respectively. The adsorption capacity of HTC-RAC was significantly higher than both of them and reached 852.10 mg/g. However, as the hydroxyl ions increased to the maximum level in the medium, there was a decrease in the capacities of all adsorbents, and it was calculated as 221.97, 213.44, and 673.27 mg/g for CAC, C-RAC, and HTC-RAC. Approximately neutral pH was considered to be the optimum pH value for the adsorption of CAC, C-RAC, and HTC-RAC. Therefore, it was decided to carry out further experiments at the natural pH value of the solution. At a 6.41 pH value, the adsorption capacities of CAC, C-RAC, and HTC-RAC for Telon Red dye were determined to be 262.66, 255.77, and 806.33 mg/g, respectively. Since the hydrothermal carbonization method is a method that increases the carbon content of moist biomass at low temperatures and high pressure and reduces both the oxygen and hydrogen content of the biomass mainly by dehydration and decarboxylation, the adsorption capacity of HTC-RAC was quite high.

Adsorbent dose is another factor affecting the adsorption process. As the amount of adsorbent increases, the contact surface of the adsorbate increases. Thus, the interaction between the adsorbate and the adsorbent increases, providing more pollutants to be removed [67,68]. For CAC, C-RAC, and HTC-RAC, different adsorbent doses of 0.1–0.6 g/L were run in separate sets, and the removal of 100 mg/L Telon Red with a pH value of 6.41 was investigated. The studies were performed isothermally at 150 rpm in a 20 °C stirrer water bath for 1 h and presented in Fig. 5b. From Fig. 5b, it was seen that Telon Red adsorption capacities increased as the adsorbent dose was raised from 0.1 g/L to 0.6 g/L for all adsorbents. Moreover, the removal efficiencies increased from 31.31 to 99.27 % for CAC, from 15.46 to 96.4 % for C-RAC, and from 86.76 to 99.2 % for HTC-RAC. Although large changes in the removal efficiency were obtained for CAC and C-RAC, a maximum removal performance was achieved for HTC-RAC at the lowest adsorbent dose. The adsorption capacities of CAC and C-RAC increased rapidly up to 0.35 g/L of adsorbent dose and achieved 262.66 and 255.77 mg/g, respectively, and then there were no significant increases. And the removal efficiencies in 0.35 mg/L dose were calculated as 90.57 % for CAC and 88.19 % for C-RAC. In Telon Red adsorption by HTC-RAC, the maximum adsorption capacity was calculated as 806.33 mg/g at a dose of 0.1 g/L. Therefore, the optimum adsorbent doses were selected as 0.35 g/L for CAC and C-RAC and 0.1 g/L for HTC-RAC, and these doses were used in subsequent experiments. In adsorption processes, choosing an adsorbent that provides high removal efficiency in low amounts is very important. As a result of the adsorption experiments, it was observed that the adsorbent dose could be significantly reduced using adsorbent obtained from the hydrothermal carbonization method.

Contact time is another factor affecting the adsorption process. In the initial stages of the adsorption process, the adsorbate rapidly enters the empty pores of the adsorbent. As the process continues, the adsorption rate decreases as adsorbate molecules accumulate on the adsorbent surface. The desorption rate becomes equal to the adsorption rate at the equilibrium point. As a result, this shows that the process is in equilibrium and the adsorption will not change with the time factor. To determine and compare the equilibrium times of all adsorbents, the

studies were performed with contact times ranging from 5 to 180 min isothermally at 150 rpm in a 20 °C stirrer water bath. The determined optimal doses of adsorbent were used to remove 100 mg/L of Telon Red at a pH value of 6.41. The amount of substance removed per unit adsorbent (q_t) was calculated for all adsorbents. The comparative plot was given in Fig. 5c. As seen in Fig. 5c, the adsorption capacities of all adsorbents increased in the first 60 min, and were calculated as 262.66 mg/g for CAC, 255.77 mg/g for C-RAC, and 806.33 mg/g for HTC-RAC. As the contact time increased, the adsorption rate of all adsorbents decreased. At the 60th minute, the removal efficiencies of CAC, C-RAC, and HTC-RAC were 90.57, 88.2 and 85.39 %, respectively, and increased to 94.81, 90.21 and 89.64 % at the 180th minute. Adsorption capacities at the 180th minute were calculated as 274.95 mg/g for CAC, 261.61 mg/g for C-RAC, and 909.82 mg/g for HTC-RAC. Since there was no significant increase in Telon Red removal efficiency after the 60th minute for all adsorbents, 60 min was determined as the equilibrium contact time. Moreover, when the optimal adsorbent doses were applied, HTC-RAC's adsorption capacity was significantly higher than that of the others, even though the equilibrium time remained unchanged.

The *initial concentration* of the pollutant changes the number of molecules adsorbed. Telon Red removal efficiencies of CAC, C-RAC, and HTC-RAC were evaluated at different initial dye concentrations. Studies were carried out isothermally at 150 rpm in a 20 °C stirred water bath at initial dye concentrations ranging from 50 to 1000 mg/L with a pH of 6.41. The adsorption performances of the optimum adsorbent doses in 60 min were determined. As seen in Fig. 5d, the dyestuff molecules at lower Telon Red concentrations came into contact with the binding sites on the adsorbents easily, and the removal efficiency for all adsorbents was more successful at the lowest dyestuff concentration than in other cases. Telon Red removal efficiencies for CAC, C-RAC, and HTC-RAC were calculated as 99.37, 93.65 and 87.32 %, respectively. The adsorption capacities were 134.11 mg/g for CAC, 126.4 mg/g for C-RAC, and 412.5 mg/g for HTC-RAC. However, due to the limited binding surfaces of adsorbents, the higher Telon Red concentration decreased the adsorption performance. At the highest dyestuff content, the removal efficiencies decreased to 15.64, 17.95, and 16.35 % for CAC, RAC, and HTC-RAC, respectively. Adsorption capacities were 454.65 mg/g for CAC, 521.91 mg/g for C-RAC, and 1664.30 mg/g for HTC-RAC. Moreover, the adsorbent created through the hydrothermal carbonization process demonstrated excellent adsorption performance even at a significantly lower adsorbent dose compared to the others. The increase in concentration creates a stronger driving force that facilitates the transfer of Telon Red dye molecules to the adsorbent. This situation increases the possibility of contact between the molecules and the active sites of the adsorbent, thus increasing the adsorption capacity [69]. According to these results, it was revealed that the initial concentration of wastewater containing dyestuff should be reduced by applying different pre-treatments to increase the adsorption performance before it is given to the adsorption units within a treatment plant.

3.2. Kinetics of Telon Red Adsorption

Adsorption kinetics provides information about adsorption rate, adsorbent performance, and mass transfer pathways. Numerous mathematical models, generally based on reaction kinetics, mass transfer diffusion, and reaction diffusion, have been proposed to explain adsorption kinetic data. The adsorption mass transfer kinetics are based on the four consecutive steps: bulk diffusion, external diffusion/fluid film diffusion, internal/intraparticle diffusion, and adsorption (physical or chemical). The first and the final step are very fast in these four consecutive steps. Therefore, the overall speed of the process is determined by the second and/or third steps. The main control step for the total rate of adsorption is the step where the resistance is greatest. The stronger the resistance at a specific phase, the greater the concentration difference needed to overcome this resistance [54,70,71].

The adsorption behaviors of CAC, C-RAC, and HTC-RAC in Telon Red removal were evaluated through adsorption kinetics. The kinetic parameters of CAC, C-RAC, and HTC-RAC for Telon Red removal were calculated after determining the adsorption velocities of each adsorbent. The Elovich model, pseudo-first-order and second-order kinetic models from adsorption reaction models, and the intraparticle diffusion model from adsorption diffusion models were analyzed to provide information about adsorption rate, adsorbent performance, and mass transfer pathways. Model parameters and R-squared values were calculated using the equations obtained from the graphs (Fig. 6). According to the determined R-squared values of the models, although the fit with other applied models was good, the second-order kinetic model provided the best fit with values >0.99 for CAC, C-RAC, and HTC-RAC (Table 1). Similar results were observed in many studies [14,17,19]. The pseudo-second-order reaction model assumes that there is an electron exchange between adsorbate and adsorbent; therefore, chemical adsorption is dominant in the process. Moreover, for each adsorbent, the equilibrium adsorption capacities calculated from the model and the equilibrium adsorption capacities obtained from the adsorption experiments were determined to be closer to each other. This indicated that the decisive step in the adsorption processes were chemisorption, not mass transfer [52].

The transfer of adsorbate to adsorbent occurs in three stages: film diffusion, intraparticle diffusion and adsorption. The intra-particle diffusion model was tested to determine the diffusion mechanism of Telon Red into adsorbents. If there is multi-linearity in the graph created from experimental data, it means that more than one step affects the adsorption process. As seen in Fig. 6, there was evidence of multi-linearity (two steps) in the graphs derived from the experimental data for each adsorbent. Table 1 also indicated that HTC-RAC had the highest intercept (C) values, but the regression line did not pass through the origin ($C \neq 0$). The deviations from the origin indicated that film diffusion and intra-particle diffusion were effective in the Telon Red adsorption of CAC, C-RAC, and HTC-RAC.

3.3. Isotherms of Telon Red adsorption

The most suitable and comprehensive approach to studying adsorption mechanisms, maximum adsorption capacity, and the characteristics of adsorbents is to simulate equilibrium adsorption data using isotherm models. Furthermore, isotherm models provide crucial input parameters for process models, such as fixed bed and solute transport models [70]. Adsorption isotherms characterize adsorption processes completely and comprehensively. The adsorption isotherm identifies and estimates the amount of adsorbate adsorbed at a constant temperature as a function of concentration or pressure [72–74]. The value of the linear regression constant (R^2) helps determine which adsorption isotherm best fits the adsorption behavior. Because it evaluates the adsorption system, quantifies the adsorbate distribution, and validates

the theoretical adsorption isotherm assumptions, the R^2 value is the most trustworthy tool for identifying the best-fitted model [54]. In this study, Langmuir, Freundlich, and Tempkin isotherm models were used to characterize the Telon Red adsorption mechanism of CAC, C-RAC, and HTC-RAC. The data collected from the experiments carried out at concentrations varying between 50 and 1000 mg/L at ambient temperatures of 20, 40, and 60 °C for all adsorbents were used in the mathematical formulas of the isotherm models, and isotherm curves were drawn. The linear regression constants and model constants were calculated from the related graphs and given in Table 2.

Based on the correlation coefficients, the Langmuir isotherm model demonstrated the linear profile with the maximum value of R^2 for the Telon Red adsorption on CAC, C-RAC, and HTC-RAC for all temperatures. As seen in Table 2, the adsorption capacity was predicted to be 1111.11 mg/g for C-RAC at 60 °C with 0.98 of R^2 value. On the other hand, the adsorption capabilities were anticipated to be 1666.67 mg/g for HTC-RAC at 20 °C with 0.99 of R^2 value, but they were found to be 666.67 mg/g for CAC at the same temperature with an R^2 value of 0.99. Moreover, the separation factor (R_L) procures knowledge related to the kind of adsorption: unfavorable ($R_L > 1$), linear ($R_L = 1$), irreversible ($R_L = 0$), or favorable ($0 < R_L < 1$). For all adsorbents, R_L values were found between 0 and 1 in all experiments performed at different temperatures and concentrations. It showed that Telon Red ions favored connecting to the surfaces of CAC, C-RAC, and HTC-RAC, and the adsorption kinds were favorable. The Langmuir adsorption isotherm model is based on four basic assumptions: monolayer adsorption, homogeneous adsorption site distribution, constant adsorption energy, and minimum adsorbate-molecule interaction [70,73,75].

Based on the findings, the adsorption of Telon Red indicated that the surfaces of CAC, C-RAC, and HTC-RAC form a homogeneous monolayer. Table 3 lists the various adsorbents' adsorption capabilities for the adsorption of dyes. CAC, C-RAC, and HTC-RAC performed well in the removal of Telon Red among the other adsorbents. The adsorbents produced from *Phragmites australis* plants by carbonization and hydro-thermal carbonization techniques were shown to have comparable adsorption capacities to various alternative adsorbents, as shown in Table 3. It follows that the produced biochars may be used as a substitute material for other materials during the adsorption process.

3.4. Temperature impacts and thermodynamics

Lu et al. examined the effects of pressure and temperature on the gas adsorption process, and indicated that temperature has an effect that cannot be disregarded, particularly when extrapolating experimental findings to underground settings [78]. Similarly, Wang et al. suggested that it is important to take into account the temperature dependence of adsorption, particularly when thinking about thermal stimulation methods for shale formations [79]. Telon Red removal efficiencies of CAC, C-RAC, and HTC-RAC were evaluated at different initial dye

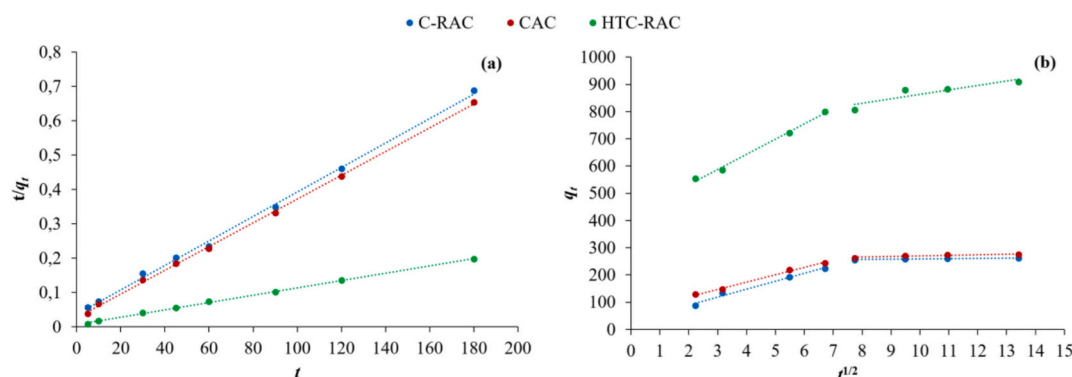


Fig. 6. Pseudo-second order reaction (a) and intra-particle diffusion (b) kinetic models of Telon Red adsorption on CAC, C-RAC, and HTC-RAC.

Table 1

The constants related to the kinetic models.

	Pseudo first order				Pseudo second order			
	$q_e \text{ exp}$	k_1	q_e	R^2	K_2	H	q_e	R^2
CAC	260.08	0.0498	213.90	0.99	0.00046	37.18	285.71	0.99
C-RAC	272.78	0.0554	255.76	0.94	0.00035	26.88	277.78	0.99
HTC-RAC	880.90	0.0612	777.67	0.84	0.00017	136.99	909.09	0.99

	Elovich						Intraparticle diffusion			
	$q_e \text{ exp}$	α	β	R^2	$k_{id,1}$	C_1	R_1^2	$k_{id,2}$	C_2	R_2^2
CAC	260.08	152.84	0.022	0.95	27.01	65.91	0.99	2.08	248.82	0.82
C-RAC	272.78	69.06	0.019	0.95	29.14	31.10	0.98	1.05	248.07	0.93
HTC-RAC	880.90	3501.70	0.009	0.98	55.95	419.90	0.99	16.53	697.75	0.81

Table 2

The constants related to the isotherm models.

	T (°C)	Langmuir			Freundlich			Tempkin		
		$q_m \text{ mg/g}$	$K_L \text{ L/mg}$	R^2	$K_F \text{ mg/g}$	$1/n$	R^2	$A_T \text{ L/g}$	$B \text{ J/mol}$	R^2
C-RAC	20	666.66	0.012	0.91	100.995	0.26	0.92	1.018	83.88	0.77
	40	769.23	0.100	0.96	150.661	0.27	0.91	1.126	110.83	0.86
	60	1111.11	0.196	0.98	159.184	0.35	0.93	1.455	186.09	0.89
CAC	20	666.67	0.093	0.99	155.346	0.14	0.93	74.634	35.83	0.83
	40	333.33	0.047	0.99	–	–	0.63	–	–	0.42
	60	158.73	0.017	0.94	–	–	0.48	–	–	0.30
HTC-RAC	20	1666.67	0.500	0.99	–	–	0.63	–	–	0.54
	40	2000.00	0.128	0.98	–	–	0.38	–	–	0.23
	60	2500.00	0.077	0.87	–	–	0.54	–	–	0.44

Table 3

A comparison of Telon Red adsorption capacities on different adsorbents.

Adsorbent	$q_m \text{ (mg/g)}$	Reference
Bis-imidazolium modified bentonite	102.57	[50]
Poly(N-vinylpyrrolidone) modified bentonite	24.60	[76]
Diphosphonium ion-exchanged montmorillonite	84.00	[77]
CAC	262.66	This study
C-RAC	255.77	This study
HTC-RAC	806.33	This study

concentrations and temperatures. Studies were carried out isothermally at 150 rpm in a 20, 40, and 60 °C stirred water bath at initial dye concentrations ranging from 50 to 1000 mg/L with a pH of 6.41. The adsorption performances of the optimum adsorbent doses in 60 min were determined.

The impacts of both temperature and the initial Telon Red concentrations on the adsorption processes were given together in Fig. 7. As seen in Fig. 7a, in Telon Red adsorption of CAC, the temperature increase had a negative impact on dye removal at all initial dye concentrations. Removal efficiency with decreasing dye concentration increased from 15.64 to 99.37 at 20 °C, from 14.06 to 95.50 % at 40 °C, and from 12.77 to 86.23 % at 60 °C. However, at the lowest dye concentration (47.24 mg/L), the removal efficiency decreased from 99.37 to 86.23 % as the temperature increased from 20 to 60 °C. From Fig. 7b, in Telon Red adsorption of C-RAC, it was observed that the temperature increase had a positive effect on dye removal at all initial dye concentrations. Removal efficiency with decreasing dye concentration increased from 17.93 to 94.11 at 20 °C, from 24.39 to 96.89 % at 40 °C, and from 36.15 to 100 % at 60 °C. However, at the lowest dye concentration (47.24 mg/L), the removal efficiency increased from 94.11 to 100 % as the temperature increased from 20 to 60 °C. As seen in Fig. 7c, in Telon Red adsorption of HTC-RAC, similar to C-RAC, the temperature increase had a positive effect on dye removal at all initial dye concentrations. Removal efficiency with decreasing dye concentration

increased from 16.35 to 87.31 at 20 °C, from 19.51 to 87.96 % at 40 °C, and from 31.13 to 89.01 % at 60 °C. However, at the lowest dye concentration (47.24 mg/L), the removal efficiency increased from 87.31 to 89.01 % as the temperature increased from 20 to 60 °C.

However, in Telon Red adsorption onto CAC, the maximum removal performance of 99.37 % was achieved at 47.24 mg/L and 20 °C, and the adsorption capacity under these conditions was calculated as 134.11 mg/g. Additionally, it was determined that a successful process was achieved by reducing the adsorbate concentration from 95.3 to 9.79 mg/L at 20 °C with a removal efficiency of 89.73 % and the adsorption capacity reaching 244.31 mg/g. Therefore, the optimum values of temperature and initial Telon Red concentrations were determined as 20 °C and 95.3 mg/L, respectively. In Telon Red adsorption of C-RAC, increasing the temperature increased the adsorption capacity. Although the maximum performance was at 60 °C, the optimum temperature was determined as 20 °C, taking into account the energy need to increase the ambient temperature. However, 95.30 mg/L of dye concentration resulted in 215.14 mg/g adsorption capacity with 78.78 % removal efficiency at 20 °C. Therefore, the optimum values of temperature and initial Telon Red concentrations were determined as 20 °C and 95.3 mg/L, respectively. In the context of Telon Red adsorption using HTC-RAC, it was observed that increasing the temperature enhanced the adsorption capacity. Although the maximum removal efficiency was achieved at 60 °C, the optimum temperature was determined as 20 °C, considering the energy needed to increase the ambient temperature. However, HTC-RAC created a significant difference from the other two adsorbents by providing an adsorption capacity of 805.50 mg/g with a removal efficiency of 84.52 % at 20 °C at a dye concentration of 95.30 mg/L.

The basic thermodynamic parameters examined to understand the characteristics of adsorption are Gibbs free energy change (ΔG), enthalpy change (ΔH), and entropy change (ΔS). The Gibbs free energy change is described by the Gibbs-Helmholtz equation Eq. (27). ΔH , ΔS , and ΔG are state functions that depend on the initial and final states of the adsorption system. Due to the extensive properties of ΔH , ΔS , and ΔG , it is crucial to consider the amount of substance that each of the

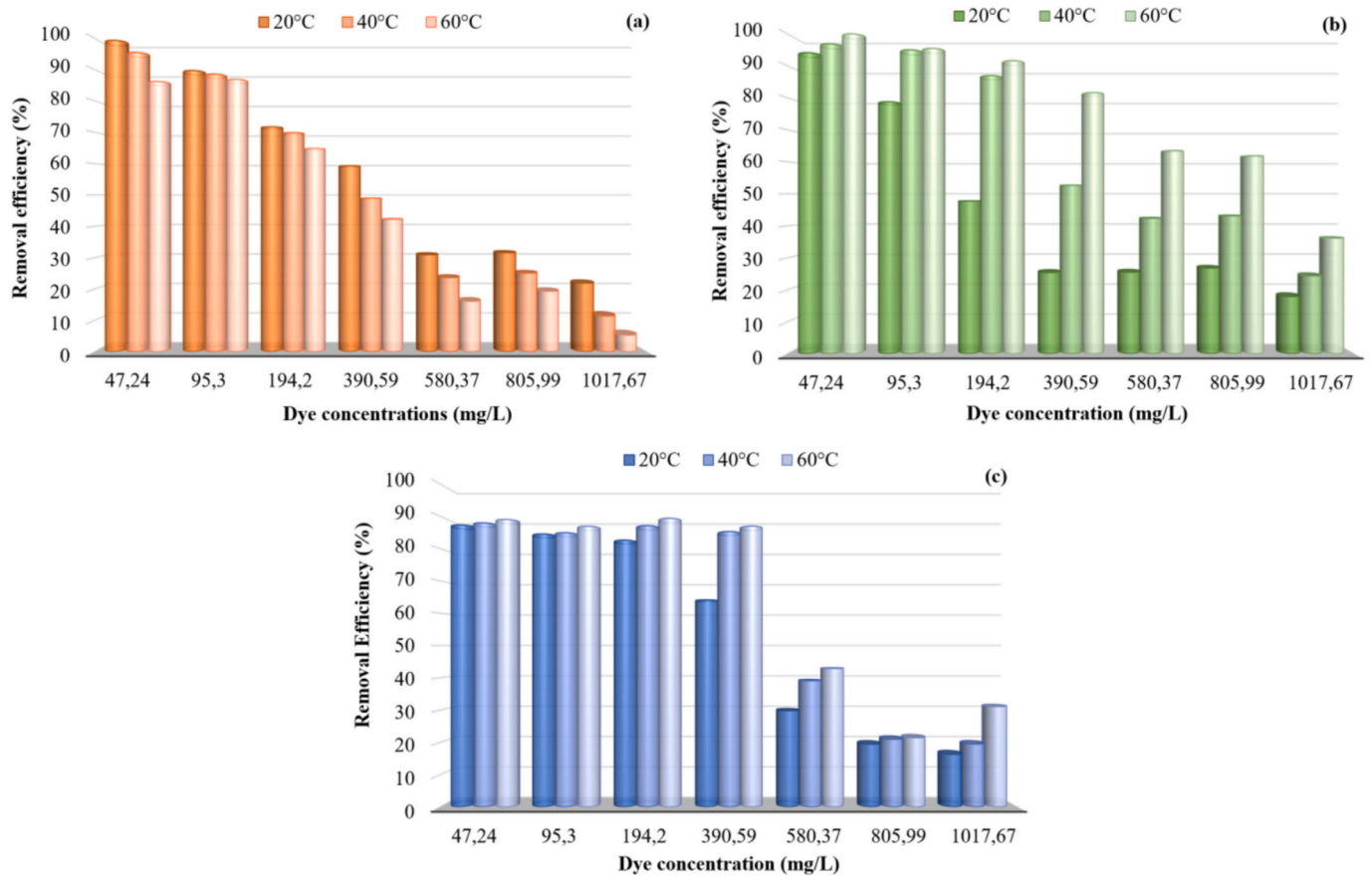


Fig. 7. The effects of temperature in Telon Red adsorption of CAC (a), C-RAC (b), and HTC-RAC (c).

thermodynamic parameters corresponds. The chemical equation of the typical adsorption reaction must be known well to examine the ΔG , ΔH , and ΔS values of adsorption (Eq. 28).

$$\Delta G = \Delta H - T\Delta S \quad (27)$$



In Eqs. 15 and 16, T is the absolute temperature (K), S is the adsorption site of the adsorbent, A is the adsorbate, $S-A$ is the adsorbate adsorbed on the adsorbent. The Van't Hoff equation defining ΔG of Eq. (28) is given for t time and equilibrium ($\Delta G = 0$, the standard conditions) in Eqs. (29) and (30), respectively. When Eqs. 27 and 30 are combined, Eq. 31 is written [80–82]:

$$\Delta G = \Delta G^\circ + RT \ln \frac{q_t/q^\circ}{(q_m - q_t)/q^\circ \times C_t/C^\circ} \quad (29)$$

$$\Delta G^\circ = -RT \ln \frac{q_e/q^\circ}{(q_m - q_e)/q^\circ \times C_e/C^\circ} = -RT \ln K^\circ \quad (30)$$

$$\ln K^\circ = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{RT} \quad (31)$$

In Eq. (29), q_t are the adsorption capacities (mg/g) of t time, C_t is the concentration of adsorbate at t time (mg/L), and R is the universal gas constant (8.314 J/mol $^\circ$ K). The maximum adsorption capacity by weight, or q_m , is the total number of adsorption sites per unit mass of adsorbent in the initial state. The amount of adsorbate adsorbed at the beginning of the process is zero. C_0 is the inlet concentration of adsorbate (mg/L); when the adsorption reaches equilibrium, the number of adsorption sites occupied by the adsorbate per unit mass of adsorbent is called the adsorption capacity and is coded by q_e . C_e is the adsorbate concentration

at the equilibrium. The $q_m - q_e$ is the number of remaining adsorption sites of adsorbent. K° is the standard equilibrium constant obtained from the isotherm model. ΔS° (J/mol $^\circ$ K) is determined by finding the point of intersection, while ΔH° (kJ/mol) is calculated by the slope of the plotted linear graph of $1/T$ versus $\ln K^\circ$.

The impacts of temperature on Telon Red removal for CAC, C-RAC, and HTC-RAC were investigated at 20, 40, and 60 $^\circ$ C. ΔH° , ΔS° , and ΔG° obtained from the Van't Hoff graphs are listed in Table 4. ΔG° is related to the feasibility and spontaneity of adsorption behavior. Gibbs free energy change, which is related to the conversion of the energy potential of a chemical reaction into work, determines the case of spontaneous occurrence of the reaction by the sign of the ΔG° value. The sign of ΔG° depends only on the numerical value of K° , and this varies according to the standard conditions chosen. When $\Delta G^\circ < 0$ of a reaction, it means that the reaction proceeds in the direction of the products until the reaction reaches the equilibrium point. As the reaction continues, ΔG° approaches zero, and when equilibrium is achieved, ΔG° is zero. Once

Table 4

ΔG° , ΔS° , and ΔH° values from dye adsorption by C-RAC, CAC, and HTC-RAC.

Adsorbent	T ($^\circ$ K)	ΔG° kJ/mol	ΔS° J/mol $^\circ$ K	ΔH° kJ/mol	R^2
CAC	293	-11,0467	-80.027	-34.669	0.98
	313	-10,0192			
	333	-779,463			
C-RAC	293	-599,156	218.99	57.685	0.94
	313	-11,9813			
	333	-14,6085			
HTC-RAC	293	-15,1388	-79.839	-38.254	0.95
	313	-12,6324			
	333	-12,0261			

the equilibrium point is passed, $\Delta G^\circ > 0$, and the reaction tends to move toward the reactants instead of spontaneously moving toward the products. The adsorption behavior is considered feasible and spontaneous when ΔG° is negative. However, positive values mean that the adsorption behavior is not spontaneous [81–83]. As seen in Table 4, ΔG° values varied in range from -11.047 to -7.795 kJ/mol for CAC, from -5.992 to -14.609 kJ/mol for C-RAC, and from -15.139 to -12.026 kJ/mol for HTC-RAC. The ΔG° values for the adsorption of Telon red ions on CAC, C-RAC, and HTC-RAC were negative across the experimental temperature range. This means that the adsorption processes have a thermodynamically spontaneous nature and feasibility of the adsorption process under the specified experimental conditions. In contrast to C-RAC, the negativity of ΔG° values decreased with increasing reaction temperature for CAC and HTC-RAC. Furthermore, the decreasing negativity of ΔG° with increasing temperature means the adsorption reaction is spontaneous. The decrease in ΔG° values with temperature indicates that the feasibility of adsorption improves with increasing temperature. This indicates that the spontaneity of the reaction increases with increasing temperature and that the adsorption is more favorable at higher temperatures for Telon Red dye removal using CAC and HTC-RAC [21].

ΔH° presents the knowledge for determining the kind of adsorption (physical or chemical process) [84,85]. The energy barrier that the reacting adsorbate molecules must overcome is expressed as ΔH° . A positive ΔH° value is defined as an endothermic nature, where no external energy input is required in the adsorption process [86]. The negative value of ΔH° means that the adsorption process has an exothermic nature, and is controlled by a physical mechanism [67]. The enthalpy changes for Telon Red adsorption on CAC, C-RAC, and HTC-RAC were calculated and listed in Table 4. As seen in Table 4, ΔH° values in Telon Red adsorption of CAC and HTC-RAC were calculated as -34.669 and -38.254 kJ/mol, respectively. The negative ΔH° value means that the Telon Red adsorption on C-RAC was managed by a physical mechanism and was exothermic. However, the ΔH° value of C-RAC was determined as 57.685 kJ/mol. It indicated that the adsorption processes were controlled by the chemical mechanism rather than the physical mechanism. The situation can be explained by the enhancement of the adsorption forces between the molecules of Telon Red and the active areas of the adsorbents. Thus, it was understood that the nature of the process in Telon Red adsorption of both adsorbents was exothermic.

The reaction's distinctive characteristics, such as its dissociative or associative mechanism, are evaluated by ΔS° [87]. After the adsorption process, a positive ΔS° value means that the randomness of the interface between the adsorbate and adsorbent rises, and a negative ΔS° means that the randomness of the interface decreases [18]. The ΔS° values of the Telon Red adsorption were calculated as -80.027 , 218.99 , and -79.839 J/mol $^\circ$ K for CAC, C-RAC, and HTC-RAC, respectively. It was shown that Telon Red molecules were randomly distributed on C-RAC during the adsorption processes, and the opposite was found for CAC and HTC-RAC.

3.5. Mathematical modeling of adsorption behaviors: hyper parameter tuning, prediction ability, reliability, optimization, and statistical evaluation

In the mathematical modeling process, initial concentration, initial pH, contact time, adsorbent dose, and temperature were used as model inputs, while adsorption capacity was considered the output/target for CAC, C-RAC, and HTC-RAC.

3.5.1. Hyperparameter tuning

The hyperparameters of the three ML algorithms used—GPR, SVR, and NNR—were first tuned to determine the best model and performance for each algorithm. The hyperparameters considered for each ML model are provided in the supplementary material. For this purpose, the

dataset was divided into three subsets: the training set, comprising 70 % of all data points (35 observations), and the validation and test sets, each comprising 15 % of all data points (8 observations). The models were trained on the training set for each parameter combination, and the optimal hyperparameter combination was determined using the validation set. Additionally, the test set was used to evaluate the prediction performance of the models with the best hyperparameter combination. The optimal hyperparameter combinations for the three ML models are provided in the supplementary material.

3.5.2. Prediction ability

3.5.2.1. Holistic evaluation. The prediction ability of the models with the best hyperparameter combinations was evaluated via some performance metrics; root mean square error (RMSE) (Eq. 32), mean absolute percentage error (MAPE) (Eq. 33), median absolute percentage error (MdAPE) (Eq. 34), median absolute error (MdAE) (Eq. 35), Willmott's Index (WI) (Eq. 36) [88], and modified index of agreement (mdWI) (Eq. 37) [89].

$$RMSE = \sqrt{\frac{1}{n} \sum_{p=1}^n (Target_p - Output_p)^2}, p = 1, 2, \dots, n \quad (32)$$

$$MAPE = median \left(\left| \frac{Target_p - Output_p}{Target_p} \right| \right), p = 1, 2, \dots, n \quad (33)$$

$$MdAPE = median \left(\left| \frac{Target_p - Output_p}{Target_p} \right| \right), p = 1, 2, \dots, n \quad (34)$$

$$MdAE = median \left(|Target_p - Output_p| \right), p = 1, 2, \dots, n \quad (35)$$

$$WI = 1 - \frac{\sum_{p=1}^n |Target_p - Output_p|}{\sum_{p=1}^n |mean(Target) - Output_p| + |mean(Target) - Target_p|}, p = 1, 2, \dots, n \quad (36)$$

$$mdWI = 1 - \frac{\sum_{p=1}^n |Target_p - Output_p|^j}{\sum_{p=1}^n \left(|mean(Target) - Output_p| + |mean(Target) - Target_p| \right)^j}, p = 1, 2, \dots, n \quad (37)$$

The best hyperparameter combinations were determined based on validation sets, and in each case, the performance of the models was measured using the test set. The test set performances of the models with the best hyperparameter combinations are summarized in Table 5 using error metrics.

The findings in Table 5 were examined from a holistic and comparative perspective. It was concluded that GPR had the best prediction performance across all metrics. According to the RMSE criterion, 2.9554, 3.4778, and 12.4128 were found for C-RAC, CAC, and HTC-RAC, respectively. The MAPE criterion is calculated as 0.8134 % for C-RAC, 0.9324 % for CAC, and 0.7924 % for HTC-RAC. The MdAPE criterion was determined to be 0.2084 % for C-RAC, 1.0105 % for CAC, and 0.3428 % for HTC-RAC. According to the MdAE criterion, C-RAC, CAC, and HTC-RAC are 0.5420, 2.7264, and 2.7002, respectively. The WI criterion is 0.9882 for C-RAC, 0.9811 for CAC, and 0.9896 for HTC-RAC. The mdWI criterion was found to be 0.9998 for C-RAC, 0.9995 for CAC, and 0.9998 for HTC-RAC. A deeper analysis of the findings suggests that the GPR modeling tool is particularly well adapted to the data

Table 5

The testing set performances of the models for the best hyperparameter.

Response parameter	Model	Metrics					
		RMSE	MAPE	MdAPE	MdAE	WI	MdWI
C-RAC	GPR	2.9554	0.8134 %	0.2084 %	0.5420	0.9882	0.9998
	SVR	46.4240	7.5647 %	7.5564 %	17.2544	0.8735	0.9829
	NNR	12.0760	4.6357 %	4.5083 %	8.3762	0.9344	0.9977
C-AC	GPR	3.4778	0.9324 %	1.0105 %	2.7264	0.9811	0.9995
	SVR	43.9895	6.0339 %	3.5369 %	11.2506	0.8785	0.9666
	NNR	16.6377	4.5706 %	3.8959 %	9.3527	0.8879	0.9875
HT-C-RAC	GPR	12.4128	0.7924 %	0.3428 %	2.7002	0.9896	0.9998
	SVR	233.0659	9.1552 %	7.8733 %	50.1608	0.8474	0.9504
	NNR	33.0576	2.7667 %	2.0586 %	30.5387	0.9763	0.9994

behavior, as it produces predictions with percentage errors around or below 1 %. NNR demonstrated strong performance and ranked second overall, with a median percentage error of 2 % for HTC-RAC. It also exhibited acceptable prediction accuracy, albeit second-best again, with an average error of approximately 4 % for C-RAC and CAC. In contrast, SVR performed the worst, with an average percentage error ranging from 5 % to 10 % for all three response parameters, raising concerns about its suitability for modeling this data process.

3.5.2.2. Individual evaluation — validity. At this stage, following a holistic and comparative evaluation, the individual performance of the most suitable modeling tool—GPR—for modeling the adsorption process, i.e., the best simulation of the process, is analyzed in detail. The corresponding visuals, presenting the compatibility of the predictions with the actual values, are presented in Fig. 8 by violin graphs consisting of a combination of a box plot and density plot, providing information about both summary statistics (median, and quartiles) and the probability density of the data. According to the violin graphs, the data points from all three adsorption processes and the predictions generated by GPR exhibit strong compatibility in terms of distribution shape and spread across all three subsets—training, validation, and testing. This is evidence of high statistical compatibility between data points.

The performance of the GPR models, as the best simulations of each of the processes, can be also demonstrated by constructing a simple linear regression model (S-LRM) between the predictions and the real observations. The S-LRM to be created is designed so that the predictions represent the independent variable ($\hat{Y}_p$) and the actual observations represent the dependent variable (Y_p). Such a S-LRM is given in Eq. 38.

$$Y_p = \beta \hat{Y}_p + \varepsilon_p, p = 1, 2, \dots, n \quad (38)$$

The basic expectation here is for the regression (β) and determination (R^2) coefficients related to the model to have a value close to 1. When coefficients take the value of 1, it indicates perfect accord between the model's predictions and the actual values. In addition, the fact that the confidence interval of the regression coefficient includes the value 1 indicates that this coefficient can take its value with a probability of 95 %. A summary of the S-LRMs generated between the predictions of the GPR models and the actual observations was given in Table 6.

From Table 6, it becomes evident that the regression and determination coefficients for all datasets in each of the three processes are nearly 1. Additionally, all confidence intervals include 1, except for the one related to the validation sub-set in the C-RAC process. This demonstrates that the predictions generated closely match the actual observed values. In other words, GPR models exhibit remarkable

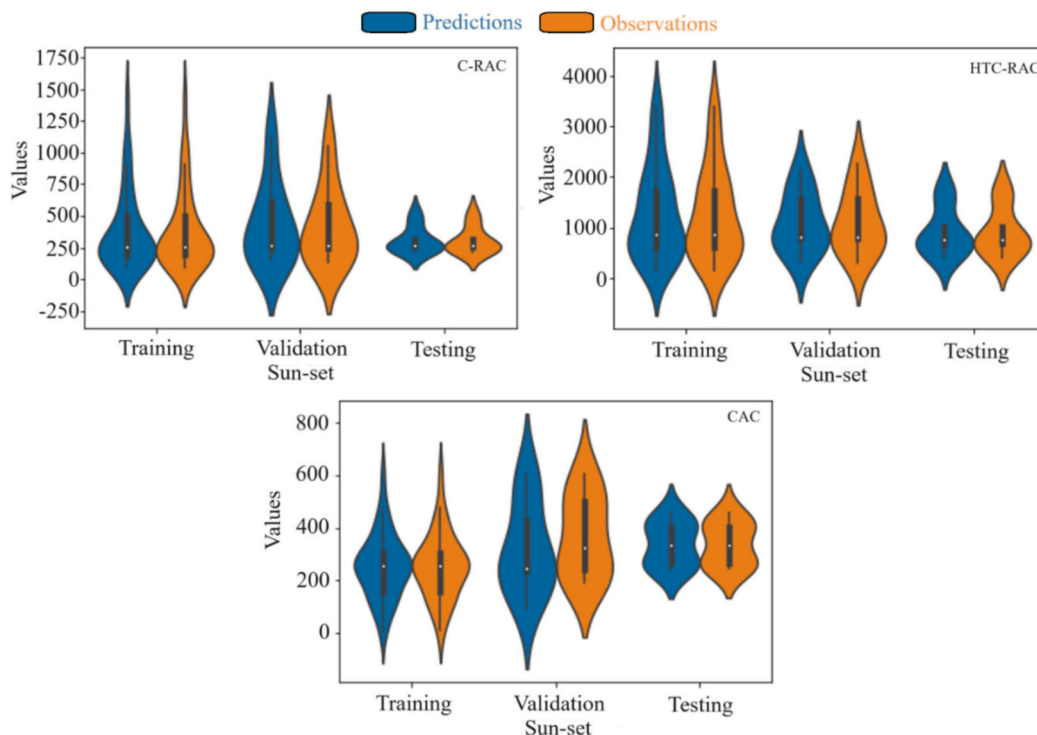
**Fig. 8.** The compatibility of the predictions of GPR with the actual values — violin graphs.

Table 6

The summary of the S-LRMs.

Response parameter	Sub-set	$Y = \hat{\beta}Y_{pre}$	95 % confidence interval of β		R^2 %
			Lower bound	Upper bound	
C-RAC	Training	$Y = 1.000072Y_{pre}$	0.998583	1.001561	99.9982
	Validation	$Y = 0.956409Y_{pre}$	0.931326	0.981492	99.9140
	Testing	$Y = 0.995603Y_{pre}$	0.988495	1.002711	99.9936
	ALL	$Y = 0.991051Y_{pre}$	0.984864	1.026984	99.9517
C-AC	Training	$Y = 1.000435Y_{pre}$	0.995167	1.005702	99.9772
	Validation	$Y = 1.022481Y_{pre}$	0.669628	1.375334	87.0253
	Testing	$Y = 1.000190Y_{pre}$	0.991228	1.009154	99.9899
	ALL	$Y = 1.005182Y_{pre}$	0.952692	1.057671	96.7310
HT-C-RAC	Training	$Y = 1.000067Y_{pre}$	0.999869	1.000265	99.9999
	Validation	$Y = 1.020031Y_{pre}$	0.989044	1.051018	99.8846
	Testing	$Y = 1.006452Y_{pre}$	0.997281	1.015622	99.9896
	ALL	$Y = 1.002849Y_{pre}$	0.998965	1.006734	99.9814

predictive performance across all three processes and can be considered *valid simulations* of them. This remarkable alignment between the predictions and the actual observed values is evident in Fig. 9, which

presents a scatter plot of predictions versus actual values.

3.5.2.3. Individual evaluation — consistency/reliability. In the previous subsection of the article, the validity of the most appropriate modeling tool, the GPR, as an evaluation of individual performance, was discussed in depth. This section, for GPR, will assess the consistency and reliability, the other two crucial characteristics desired in any modeling tool or simulation of a process. In this respect, the GPR was run 50 times with the best hyperparameter combinations for each of the three processes. The results were evaluated using visual presentations and statistics based on the MdAPE criterion obtained for test subsets in each process. The statistics of MdAPE criterion values obtained from 50 runs were determined (see supplementary material). The MdAPE criterion becomes evident that even the maximum MdAPE values for all three processes are relatively small. While these values are approximately 1.5 % for CAC, they are even lower for C-RAC (0.41 %) and HTC-RAC (0.52 %), falling below 1 %. Furthermore, the standard deviation values, indicating the variation in MdAPE values across all three adsorption processes, are notably small: 0.000977 (C-RAC), 0.002792 (CAC), and 0.001190 (HT-C-RAC). This serves as evidence of the reliability and consistency of the GPR model. The radar and box plots provided in Fig. 10 further reinforce the statistical evidence that GPR is a highly reliable and consistent modeling tool and even an applicable simulation of all three adsorption processes.

3.5.3. Process optimization via genetic algorithm

This study devised modeling tools utilizing various machine-learning algorithms to characterize three distinct adsorption processes. Moreover, the study optimized the hyperparameters of these algorithms through the use of a validation set. It identified the best modeling tool and the most realistic simulations of the processes. The validity and

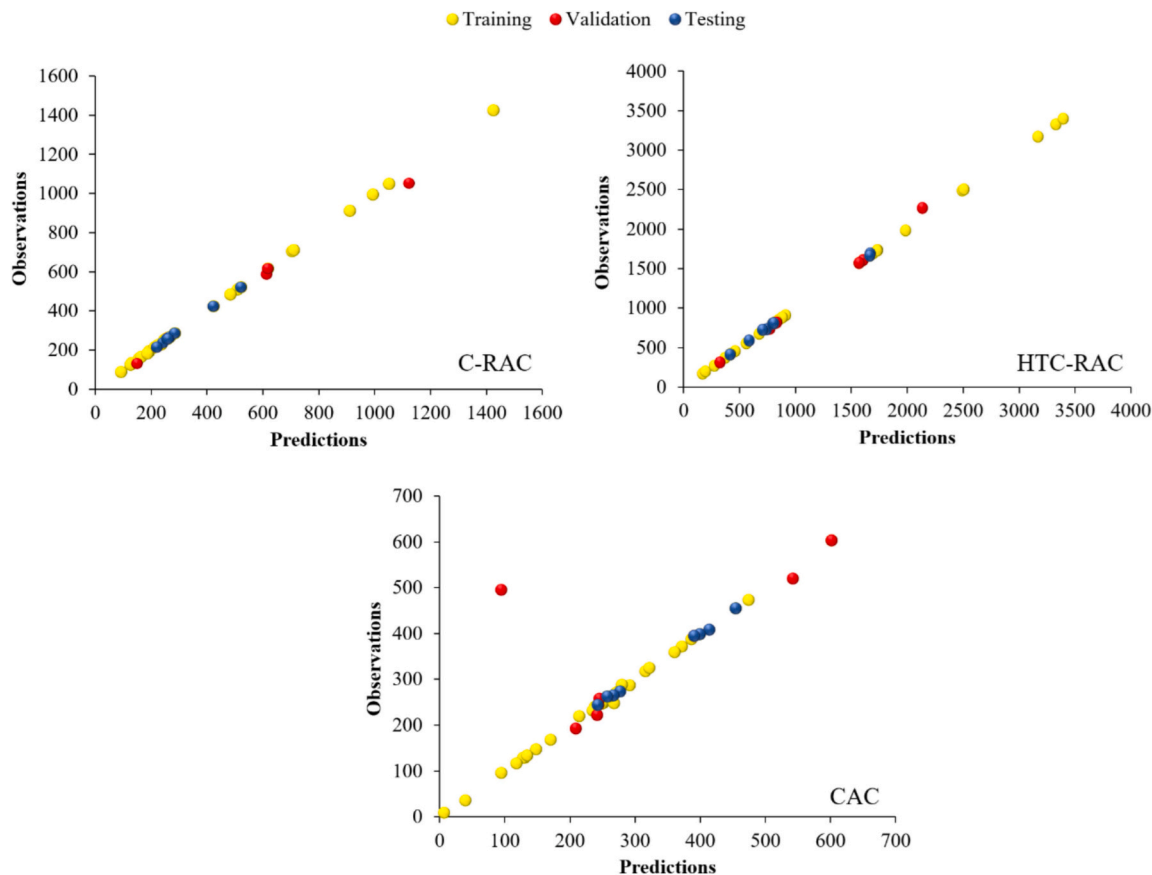


Fig. 9. The scatter plots of predictions versus actual values.

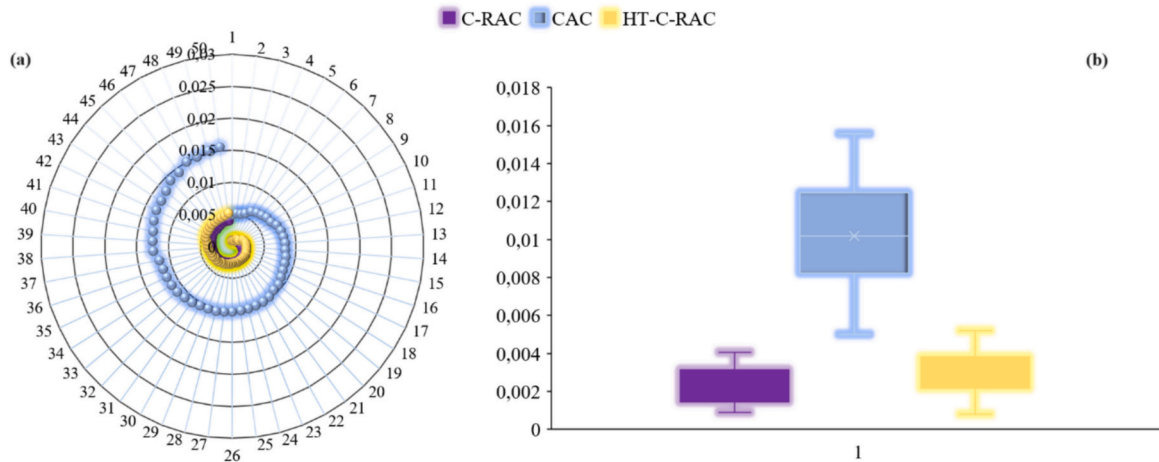


Fig. 10. The presentations of change of MdAPE values with the radar plot (a) and box plots (b).

consistency features, essential sought-after characteristics for any ML algorithm, were discussed from various angles for the chosen optimal modeling tool. Although the evaluations up to this point are very valuable, process optimization, which is invaluable for experimental design research, was also carried out in this study. So, there is still a fundamental question to be answered for modeling the adsorption process. With which values of independent variables can the best performance be achieved in the adsorption process? This means simulating which input parameter values will produce maximum adsorption performance. The answer to this question is given in this study via GA. The best simulations of all three adsorption processes, optimal GPR models, were optimized. Input parameter values that achieved the highest adsorption performance were determined with GA. In broad terms, an optimization problem with N decision variables and $J + K$ constraints can be formulated as Eq. (39). In the adsorption process, there are 5 input variables (initial concentration—IC, pH, contact time—CT, adsorbent dose—AD, temperature, T). The output/response variable was q_e . The optimization performance was evaluated using desirability coefficients as specified in Eqs. (40) and (41). The optimization results for each adsorption process are listed in Table 7.

$$\begin{aligned} & \min/\max(f(\underline{x})) \quad (\underline{x} = [x_1 \ x_2 \dots x_N]) \\ & g_j(\underline{x}) \leq 0 \quad ; \quad j = 1, 2, \dots, J \\ \text{subject to} \quad & h_k(\underline{x}) = 0 \quad ; \quad k = 1, 2, \dots, K \\ & x_i^{(L)} \leq x_i \leq x_i^{(U)} \quad ; \quad i = 1, 2, \dots, N \end{aligned} \quad (39)$$

$$d^{\max} = \begin{cases} 0 & ; \quad f(\underline{x}) < \text{Min} \\ \left(\frac{f(\underline{x}) - \text{Min}}{\text{Max} - \text{Min}} \right) & ; \quad \text{Min} < f(\underline{x}) < \text{Max} \\ 1 & ; \quad f(\underline{x}) > \text{Max} \end{cases} \quad (40)$$

$$d^{\min} = \begin{cases} 1 & ; \quad f(\underline{x}) < \text{Min} \\ \left(\frac{f(\underline{x}) - \text{Max}}{\text{Min} - \text{Max}} \right) & ; \quad \text{Min} < f(\underline{x}) < \text{Max} \\ 0 & ; \quad f(\underline{x}) > \text{Max} \end{cases} \quad (41)$$

Table 7, which presents information on process simulation optimization using GA, shows that nearly 100 % desirability was achieved for all three processes. This result is invaluable when designing new experiments is costly and time-consuming. Because, these results indicate which input parameter values will yield the highest efficiency in any adsorption process, eliminating the need for further experimentation.

3.5.4. Statistical evaluation

As part of mathematical modeling, the adsorption performances (QE values) obtained for all three processes were statistically examined to determine if there were any differences between them. For this purpose, a one-way analysis of variance was conducted on the output values of all three processes, totaling 51 observations. For this, first of all, the output values of all three processes must satisfy the assumption of normal distribution. The Kolmogorov-Smirnov normality test, typically recommended for datasets with >30 observations, was employed to assess the assumption of normal distribution (Table 8). As seen in Table 8, it

Table 7
The results of the optimization.

Process	Decision variable	Constraint	Optimal value	Objective function value	Max _{observed}	Desirability
C-RAC	IC	$47.24 \leq IC \leq 1017.67$	823.5827	1441.6164	1425.6857	100.00 %
	pH	$3.01 \leq pH \leq 11.12$	4.7929			
	CT	$5.00 \leq CT \leq 180$	122.4369			
	AD	$0.10 \leq AD \leq 0.60$	0.3732			
	T	$20.00 \leq T \leq 60.00$	58.8890			
CAC	IC	$47.24 \leq IC \leq 1017.67$	823.5827	606.7822	602.9429	100.00 %
	pH	$3.01 \leq pH \leq 11.12$	8.7134			
	CT	$5.00 \leq CT \leq 180$	45.6459			
	AD	$0.10 \leq AD \leq 0.60$	0.2689			
	T	$20.00 \leq T \leq 60.00$	21.6433			
HTC-RAC	IC	$47.24 \leq IC \leq 1017.67$	391.0553	3389.3214	3395.6000	99.81 %
	pH	$3.01 \leq pH \leq 11.12$	6.9972			
	CT	$5.00 \leq CT \leq 180$	73.9135			
	AD	$0.10 \leq AD \leq 0.60$	0.1105			
	T	$20.00 \leq T \leq 60.00$	58.8932			

Table 8

The results of the statistical evaluation.

Process	Mean	95 % confidence interval for mean	Kolmogorov-Smirnov — Sig.	Kruskal Wallis test — Sig.
C-RAC	376.5887	295.9971 457.1804	<0.001	<0.001
CAC	284.7585	248.0315 321.4856	0.003	
HTC-RAC	1205.6559	973.8390 1437.4727	<0.001	

becomes evident that the distribution of output values for all three processes deviates from normality, as indicated by the respective significance values (<0.001 for C-RAC; 0.001 for CAC; <0.001 for HTC-RAC) at the 0.05 significance level. In cases where the Normal Distribution assumption is not met, the Kruskal-Wallis H test, serving as a non-parametric equivalent to one-way analysis of variance (ANOVA), should be employed. Table 8, which also presents the findings of the Kruskal-Wallis H test, indicates a statistically significant difference among the outcomes of the processes at the 0.05 significance level. Upon examining individual averages, it is evident that the outcomes of the HTC-RAC process significantly differ from the others and consistently trend higher. Looking from another angle, upon examining the confidence intervals for the averages of all three processes, it becomes apparent that the average confidence intervals of the C-RAC and CAC processes overlap. This result indicates that there is no statistically significant difference in terms of the outputs and performances between these two processes. Nevertheless, the confidence interval generated for the average of the HTC-RAC process does not overlap with those of the other two process outputs. This result shows that the HTC-RAC process differs from the other two, which indicates its tendency to produce high performance.

4. Conclusions

In this study, the adsorption behaviors of *Phragmites australis*-based biochars and commercial activated carbon were compared for Telon Red dye removal. The adsorption capacity of the adsorbent formed by the hydrothermal carbonization process was significantly higher than the others. The adsorption processes were feasible and spontaneous based on thermodynamic characteristics. Moreover, this study was designed to model the processes that reveal the adsorption capacities of CAC, C-RAC, and HTC-RAC, and to create valid and reliable simulations. By mathematical modeling, this study has successfully generated valid and consistent simulations for all three processes, as evidenced by the metrics and analyses shown from multiple perspectives. Three distinct ML algorithms were trained using various hyperparameter combinations, and the optimal combination was identified through validation sets. The optimal outcomes achieved through hyperparameter tuning revealed that GPR is the most effective modeling tool and produced the closest-to-reality simulation for all three processes. Moreover, via GA, these valid and reliable simulations are optimized to have the highest adsorption capacity with almost 100 % desirability, which is an invaluable possibility for researchers in many cases. Finally, the adsorption capacities of all three processes were compared statistically, and it was concluded that the adsorption capacity of HTC-RAC was significantly higher than the others. This suggests a promising avenue for sustainable waste management and environmental remediation, as utilizing plant-derived biochars could reduce reliance on synthetic adsorbents. Furthermore, the versatility of these biochars indicates their potential application in a wide range of wastewater treatment scenarios, enhancing their value in industrial and municipal settings.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.desal.2025.118887>.

Data availability

The authors do not have permission to share data.

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